

(Optional) : Getting Started with Google Looker Studio

Welcome to this instructional reading on getting started with Google's Looker Studio.

This instructional reading will give you the basic knowledge required to help you complete the hands-on labs and final assignment for Looker Studio if you decide to use this product instead of IBM Cognos Analytics.

Looker Studio was originally named Google Data Studio, and it is essentially a free cloud-based tool to visualize your data using a variety of visualizations, including dashboards and multipage reports.

In this reading, we will discuss:

- Navigating the Looker Studio interface
- Creating reports
- Connecting to data sources
- Adding visualizations to a report
- Configuring theme and layout settings

Navigating the Looker Studio Interface

You can access Looker Studio by navigating to <https://lookerstudio.google.com/> in your web browser. All you need to access Looker Studio is a valid free Google account.

You will be redirected to the Recent section of the Looker Studio home page, where you will see a list of templates to choose from to start your first report. You will also see any recent reports and dashboards you have already worked on.

Looker Studio

Search Looker Studio

Create

Recent Reports Data sources Explorer

Recent

Start with a Template

Blank Report Looker Studio

Tutorial Report Looker Studio

Acme Marketing Google Analytics

Search Console Report Search Console

Template Gallery

Name Owned by anyone Last opened by me Location

Create a Report

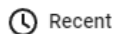
Use the Create button to add one.

Here, on the **Reports** tab, we can see that there are no recent files, as this is our first time opening the Looker Studio interface. The same applies to the **Data sources** and **Explorer** tabs, too.

The **Template Gallery** can be accessed by clicking **Templates** in the left navigation menu on the home page.



Create



Recent



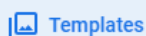
Shared with me



Owned by me



Trash

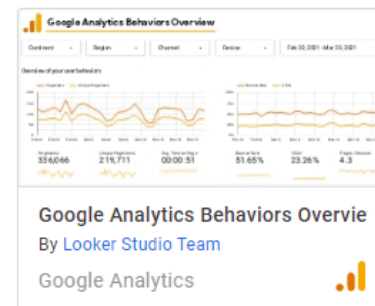
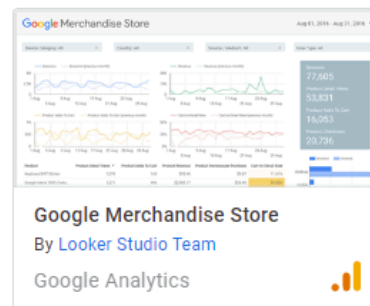
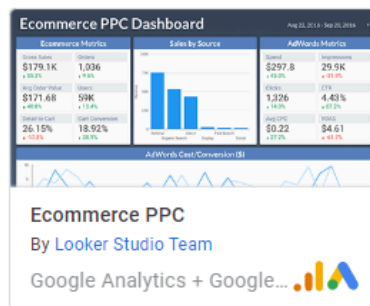
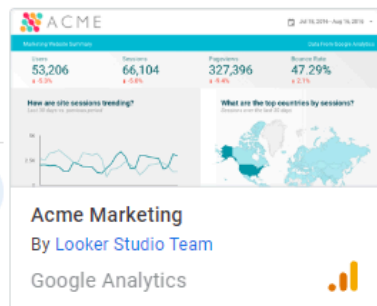


Templates

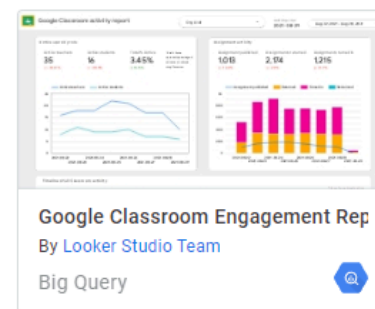
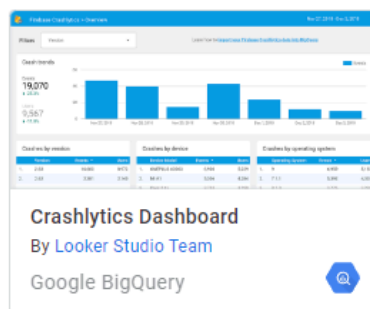
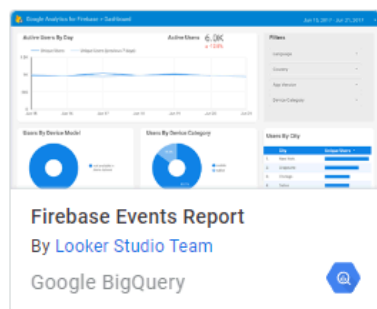
Template Gallery

Category

Google Analytics

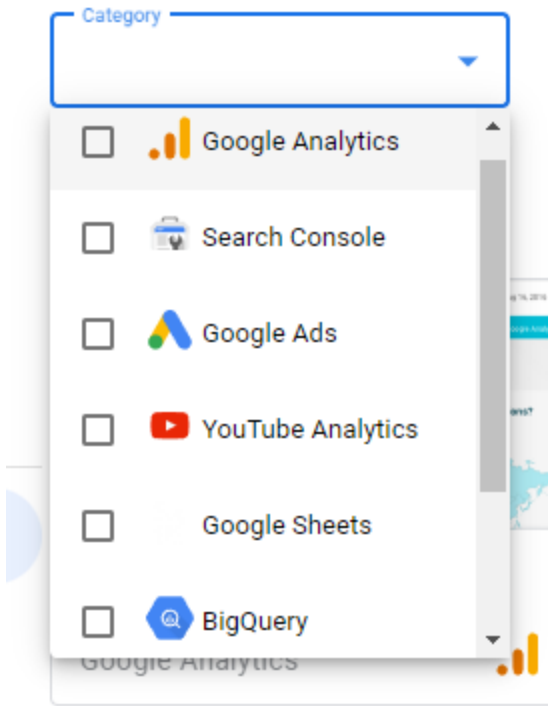


BigQuery

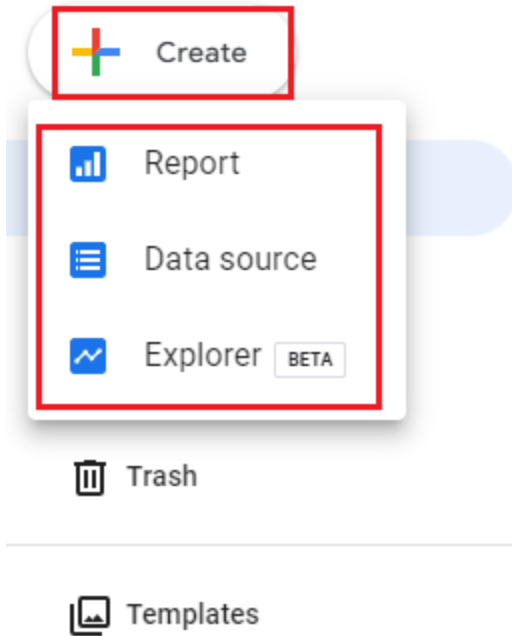
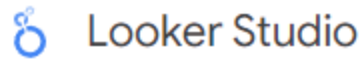


You can also browse the various templates by their category, such as Google Analytics, Google Sheets, and BigQuery, by scrolling down the page, or you can filter the Template Gallery view by one or more of those categories.

Template Gallery



From the **Create** button, you can create Reports, Data sources, and Explorers.



We will discuss **Reports** and **Data sources** in the readings in this course, and you will practice with them in the hands-on labs and the final assignment. However, you won't use **Explorers** in this course, as they are currently an experimental feature in Beta.

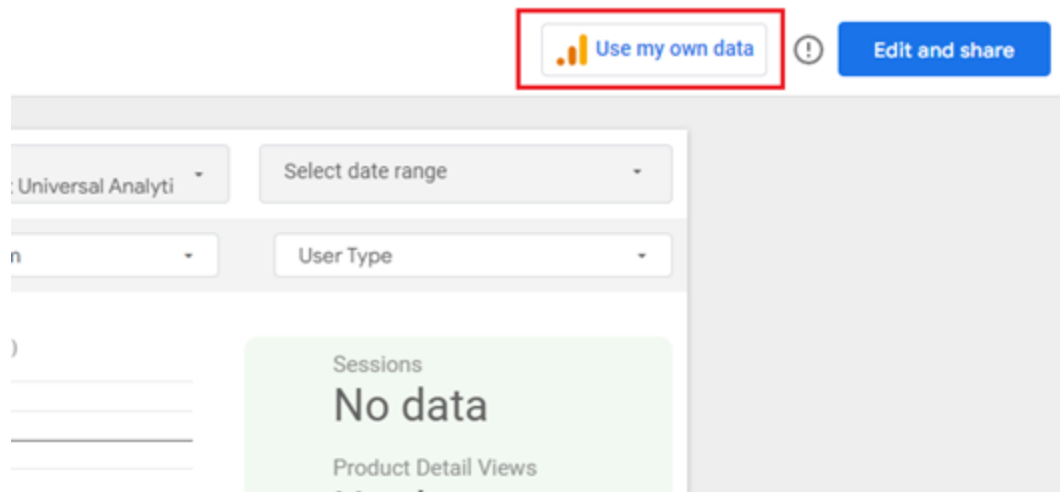
Creating Reports

You create reports in Looker Studio to enable you to visualize your data and get better insights into your data. You can then share those reports with your colleagues or customers.

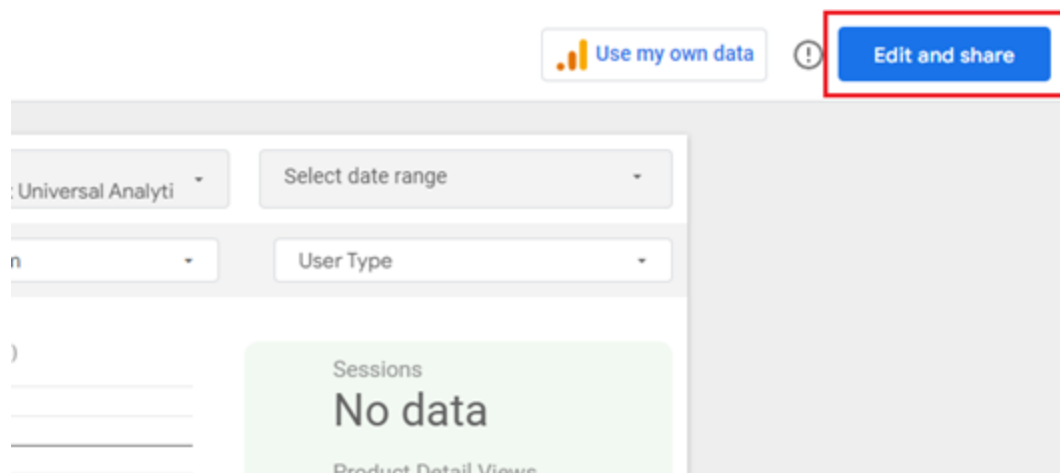
You can either create a blank report from scratch and add components to it, or you can create a report from an existing template and either use it as it is or modify it to meet your own needs.

To create a report from a template:

1. Click **Templates**.
2. Select a template from the gallery.
3. In the template, you can use the provided dataset or click **Use my own data** to upload your own data and populate the template with it.



1. You can start to edit the template by choosing **Edit and share**.



1. Click **ADD TO REPORT** in the pop-up dialog box.

You are about to add data to this report



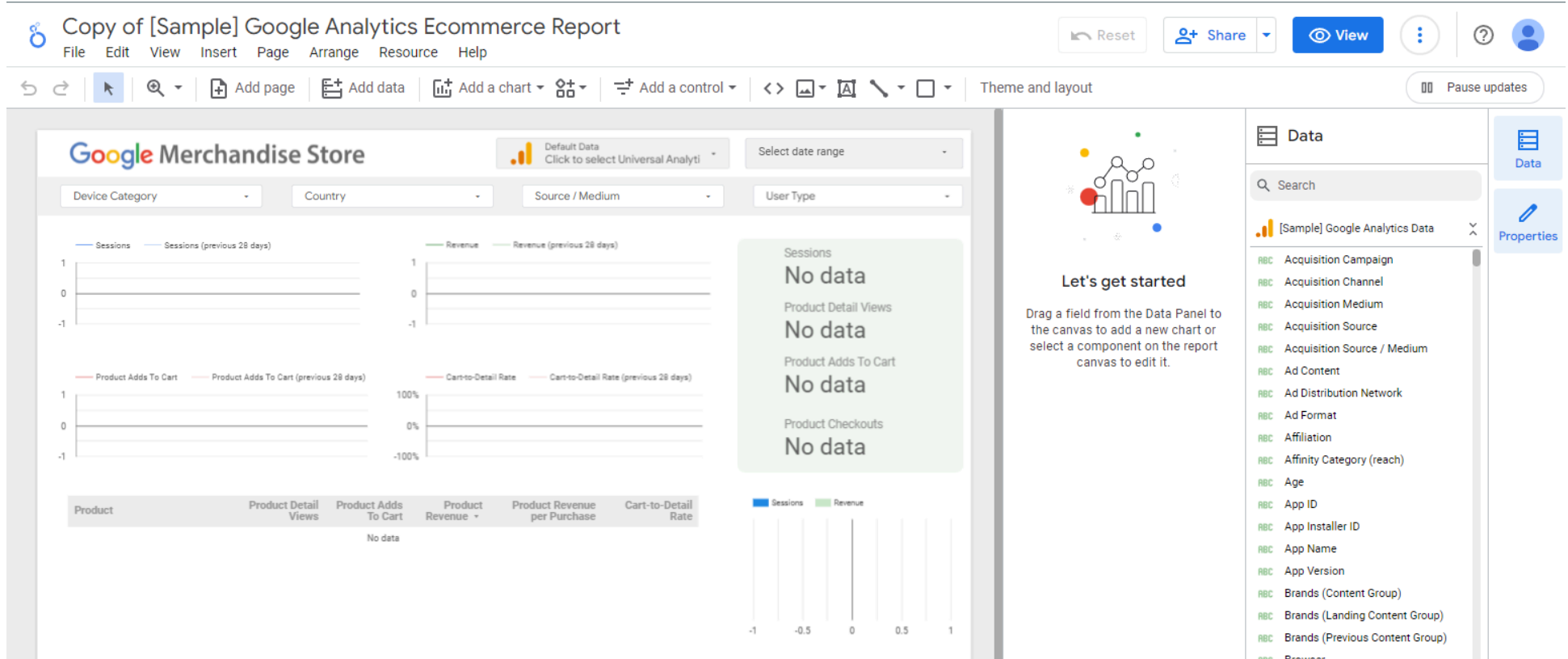
Note that **Report Editors** can create charts using the new data source(s), and can add dimensions and metrics not currently included in the report.

☐ Don't show me this again

CANCEL

ADD TO REPORT

1. Then, use the **Report Editor** to make changes to the copy report from the template.



To create a blank report:

1. Click **Create**, then select **Report**.
2. On the **Add data to report** page, you need to select a *connector* for your data source.
3. After selecting a connector, you can drag and drop your file directly onto the window.

[←](#) Add data to report

Data credentials: Steve Ryan [×](#)

File Upload
By Google
You can bring data into Looker Studio from almost any source by uploading CSV (comma-separated values) files. File upload lets you report on data not supported by a specific connector.
[LEARN MORE](#) [REPORT AN ISSUE](#)

Data Sets

cust_table.csv
cust_loyalty_table.csv
CustomerLoyaltyProgram.csv

Drag files here to create a new Data Set.

Or, if you prefer

[CLICK TO UPLOAD FILES](#)

[Cancel](#) [Add](#)

1. Alternatively, you can click the **CLICK TO UPLOAD FILES** button, then browse to the location where the dataset file is stored and open it.

[←](#) Add data to report

Data credentials: Steve Ryan [×](#)

File Upload
By Google
You can bring data into Looker Studio from almost any source by uploading CSV (comma-separated values) files. File upload lets you report on data not supported by a specific connector.
[LEARN MORE](#) [REPORT AN ISSUE](#)

Data Sets

cust_table.csv
cust_loyalty_table.csv
CustomerLoyaltyProgram.csv

Drag files here to create a new Data Set.

Or, if you prefer

[CLICK TO UPLOAD FILES](#)

[Cancel](#) [Add](#)

1. The file will start being processed.

sentiment.csv

TOTAL FILE SIZE — (0% of 100MB used)	NUMBER OF FILES 0	CREATION DATE 9/25/23 5:26 PM	LAST MODIFIED DATE 9/25/23 5:26 PM
---	----------------------	----------------------------------	---------------------------------------

ADD FILES Files must contain the same schema. [Learn More](#)

File name	Uploaded at	Size	Status
sentiment.csv	9/25/23 5:26 PM	2 MB	● Processing

1. Once the file has finished being processed and uploaded, the status will change to **Uploaded**, and you can click **Add** to add it as a data source to your blank report.

sentiment.csv

TOTAL FILE SIZE 2 MB (3% of 100MB used)	NUMBER OF FILES 1	CREATION DATE 9/25/23 5:26 PM	LAST MODIFIED DATE 9/25/23 5:26 PM
--	----------------------	----------------------------------	---------------------------------------

[VIEW FILES IN CLOUD](#)

ADD FILES Files must contain the same schema. [Learn More](#)

File name	Uploaded at	Size	Status
sentiment.csv	9/25/23 5:26 PM	2 MB	● Uploaded

[Cancel](#) [Add](#)

1. The untitled report will be displayed with the data added.

The screenshot shows the Google Looker Studio interface. At the top, there's a menu bar with 'File', 'Edit', 'View', 'Insert', 'Page', 'Arrange', 'Resource', and 'Help'. To the right are buttons for 'Reset', 'Share', 'View', and a settings icon. Below the menu is a toolbar with icons for navigation, zoom, and adding elements like pages, data, charts, and controls. The main workspace displays a table with the following data:

	Sentiment	Record Count
1.	Positive	44,885
2.	Neutral	17,216
3.	Negative	14,528

At the bottom right of the table, it says '1 - 3 / 3' with navigation arrows. On the right side, there's a sidebar with two tabs: 'Chart' and 'Data'. The 'Chart' tab is active, showing 'SETUP' and 'STYLE' sections. Under 'SETUP', 'Data source' is 'sentiment.csv', 'Date Range Dimension' is 'Date', 'Dimension' is 'Sentiment', and 'Metric' is 'Record Count'. The 'Data' tab is also visible, showing a search bar and a list of fields: 'Date', 'Postal Code', 'Sentiment', 'Year', and 'Record Count'.

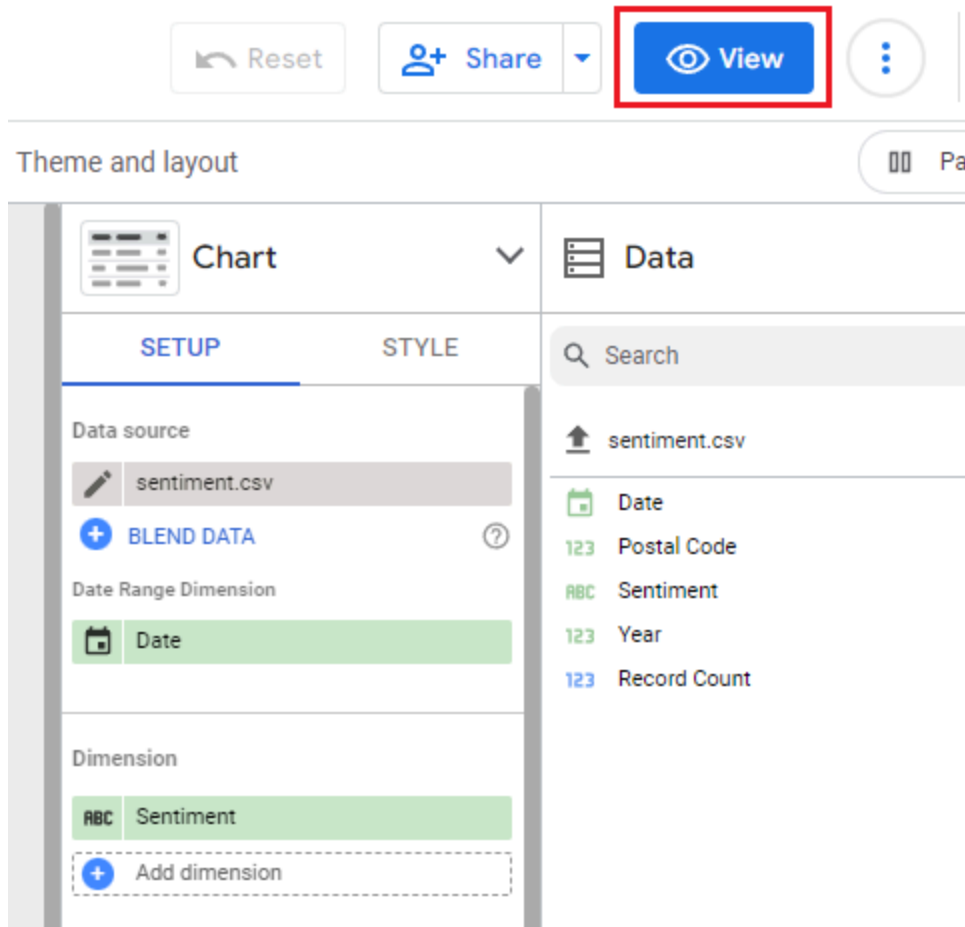
1. Looker Studio will make assumptions about how you want the data displayed and visualized, but you can modify this as you see fit and add more data fields to the report to create more visualizations from.

The screenshot shows the Google Looker Studio interface. At the top, there's a menu bar with 'File', 'Edit', 'View', 'Insert', 'Page', 'Arrange', 'Resource', and 'Help'. To the right are buttons for 'Reset', 'Share', 'View', and a menu icon. Below the menu bar is a toolbar with icons for navigation, zoom, adding pages, data, charts, controls, and theme/layout. The main workspace displays a table with the following data:

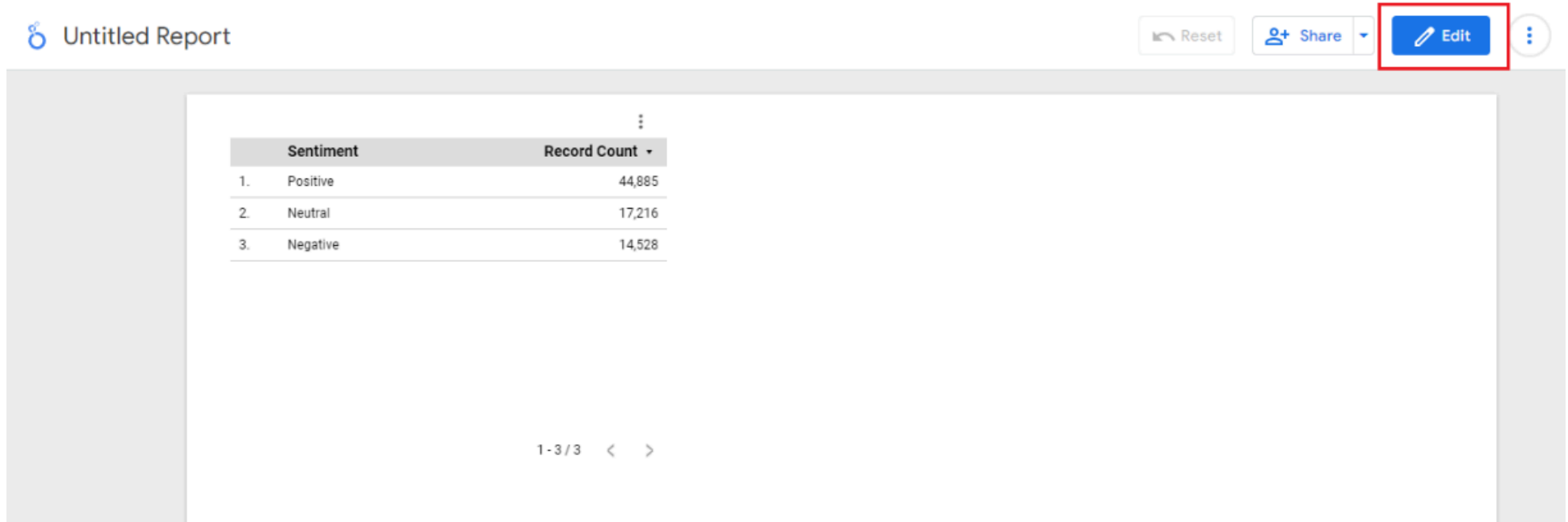
	Sentiment	Record Count
1.	Positive	44,885
2.	Neutral	17,216
3.	Negative	14,528

On the right side, there's a 'Chart' and 'Data' panel. The 'Data' panel is active, showing the 'sentiment.csv' data source. It lists dimensions: 'Date', 'Postal Code', 'Sentiment', and 'Year'. The 'Sentiment' dimension is selected. Below the dimensions, there's a 'Metric' section where 'Record Count' is selected. The 'Optional metrics' toggle is turned off.

1. Click the **View** button in the top right of the page to view the report as a user would see it.



1. The report will be displayed in read-only mode. Click the **Edit** button in the top right of the page to return to edit mode.



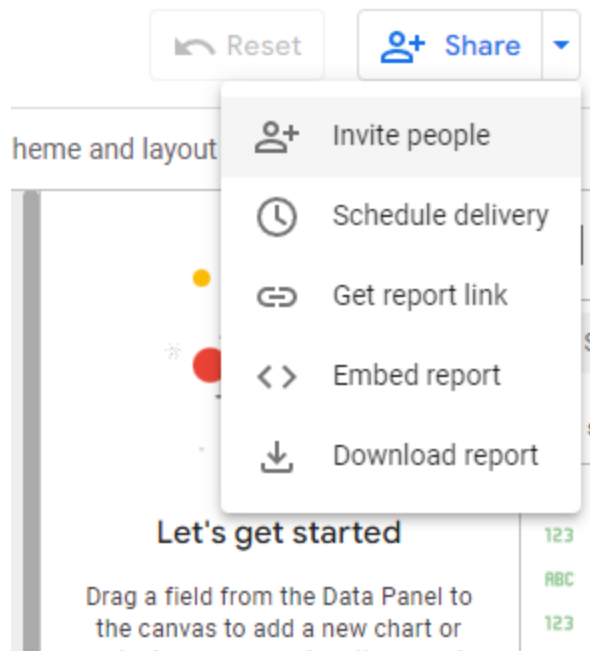
Untitled Report

Reset Share Edit

Sentiment	Record Count
1. Positive	44,885
2. Neutral	17,216
3. Negative	14,528

1 - 3 / 3 < >

1. You can use the **Share** button in either 'Edit' or 'View' mode to share your report with colleagues.



Reset Share

- Invite people
- Schedule delivery
- Get report link
- Embed report
- Download report

Let's get started

Drag a field from the Data Panel to the canvas to add a new chart or

1. The report can either be shared as an email invitation to individual users or groups of users.

Share with people and groups



Share as  Steve Ryan

Add people and groups

People with access

This content is managed by an organization and members within it

S

Owner



Viewer ▼

1. It can also be shared via a link.

Link settings



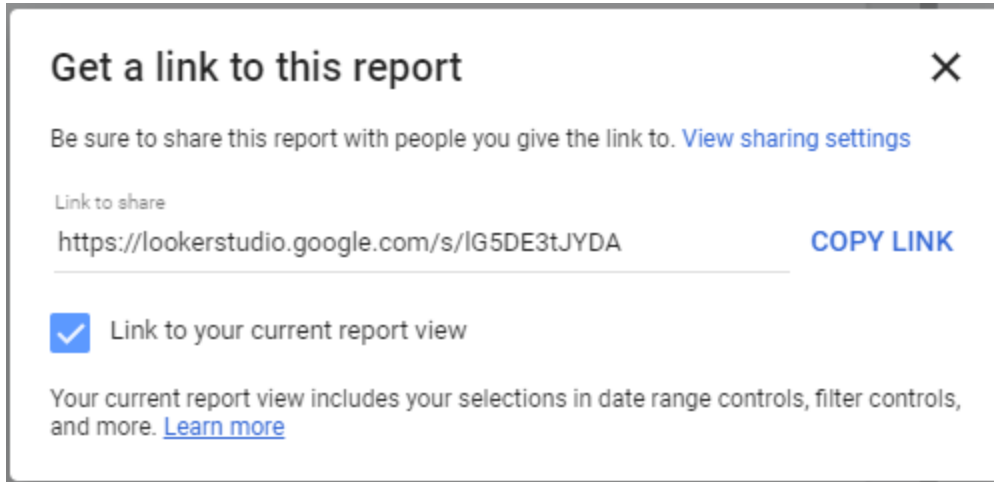
Restricted ▼

Only people added can open with this link

 Copy link

Done

1. Then, you can copy the link and send it directly to those people who you wish to be able to access it via the link.



You can also embed the report as code or as a URL or download the report as a PDF and then share it as a file instead.

Connecting to Data Sources

Data sources enable you to manage the connections to your data and configure the data fields in your data that you will use in your reports. You may also see the term *connector* being used instead of *data source*, but they are two slightly different components.

Connectors and data sources

Connectors are the components that connect your underlying data to Looker Studio features.

Whereas data sources are the components that represent a specific instance of a connector, such as a connection to a Google Sheets spreadsheet, an SQL Server database, or a CSV file.

So, *connectors* are used by a *data source* to retrieve your data from a given data platform or database product.

Whenever you create a report, you'll notice that you are first asked for a data source to connect to.

As we've seen, on the Connect to Data tab of the Add Data to Report window, you can create a new data source for your report by selecting one of many available connectors, such as the Google Sheets connector.

Add data to report

Connect to data

My data sources

Search

Google Connectors (23)

Connectors built and supported by Looker Studio [Learn more](#)

 **Looker**
By Google
Connect to your Looker semantic models.

 **Google Analytics**
By Google
Connect to Google Analytics.

 **Google Ads**
By Google
Connect to Google Ads performance report data.

 **Google Sheets**
By Google
Connect to Google Sheets.

 **BigQuery**
By Google
Connect to BigQuery tables and custom queries.

 **AppSheet**
By Google
Connect to AppSheet app data.

 **File Upload**
By Google
Connect to CSV (comma-separated values) files.

 **Amazon Redshift**
By Google
Connect to Amazon Redshift.

 **Campaign Manager 360**

 **Cloud Spanner**

 **Cloud SQL for MySQL**

 **Display & Video 360**

Then, you need to either navigate to a local file to upload or connect to a database system, or in the case of the Google Sheets connector, you need to select a worksheet from your Google Drive to add.

<https://www.coursera.org/learn/data-visualization-dashboards-excel-cognos/supplement/Nw5vE/optional-getting-started-with-google-looker-studio>

18/44

[←](#) Add data to report

Data credentials: Steve Ryan [×](#)

Google Sheets
By Google
The Google Sheets connector allows you to access data stored in a Google Sheets worksheet.
[LEARN MORE](#) [REPORT AN ISSUE](#)

ALL ITEMS	Spreadsheet	Worksheet	Options
OWNED BY ME	Copy of [template to make a copy] T&L Pre-...	Medalists	☒ Use first row as headers Column headers must be unique. Columns with empty headers will not be added to the data source.
SHARED WITH ME	[template to make a copy] T&L Pre-Launch ...	Pivot Table 1	☒ Include hidden and filtered cells
STARRED	Copy of SRYan_Course_syllabus	Country Codes	☐ Include specific range
URL	SRYan_Course_syllabus		
OPEN FROM GOOGLE DRIVE	Copy of [TEMPLATE] Course syllabus temp...		
	[TEMPLATE] Course syllabus template (for ...		
	[existing content analysis] Dartmouth - Intr...		
	Copy of Coursera Course Template		
	Week2_Store-Sales		
	Original_dashboard_webinar - Copy		
	CEO-Salary-_small-firms_		
	Height-and-Weight		
	Summer-Olympic-medals-1976-to-2008		

[Cancel](#) [Add](#)

Then, the **Report Editor** will open a new report with your selected data already embedded.

Untitled Report

File Edit View Insert Page Arrange Resource Help

Reset Share View

Add page Add data Add a chart Add a control Theme and layout

	City	Record Count
1.	Beijing	2,042
2.	Sydney	2,015
3.	Athens	1,998
4.	Atlanta	1,859
5.	Barcelona	1,705
6.	Seoul	1,546
7.	Los Angeles	1,459
8.	Moscow	1,387
9.	Moscow	1,387

1 - 9 / 9

Chart

SETUP **STYLE**

Data source

Summer-Olympic-medals-19...

BLEND DATA

Date Range Dimension

Add dimension

Dimension

City

Add dimension

Drill down

Metric

Record Count

Add metric

Optional metrics

Metric sliders

Rows per Page

100

Data

Search

Summer-Olympic-medals-1976-to-20...

Athlete

City

Country

Country_Code

Discipline

Event

Event_gender

Gender

Medal

Sport

Year

Record Count

Add a field

Alternatively, if you select the **My data sources** tab of the **Add data to report** window, you can add an existing data source to your report by selecting from one of your listed data sources.

Add data to report

Connect to data

My data sources

Q Search

Name	Owned by anyone	Last opened by me
CustomerLoyaltyProgram.csv	Steve Ryan	Sep 18, 2023
[Sample] World Population Data 2005 - 2014	Looker Studio	—
[Sample] Google Analytics Data	Looker Studio	—
[Sample] AdWords Data	Looker Studio	—
[Sample] YouTube Data	Looker Studio	—
[Sample] Rio Olympics Data	Looker Studio	—

Add

After selecting an existing data source, add it to your report.

Add data to report ✕

Connect to data [My data sources](#)

Name	Owned by anyone ▼	Last opened by me ▼ ↓
 CustomerLoyaltyProgram.csv	Steve Ryan	Sep 18, 2023
 [Sample] World Population Data 2005 - 2014	Looker Studio	—
 [Sample] Google Analytics Data	Looker Studio	—
 [Sample] AdWords Data	Looker Studio	—
 [Sample] YouTube Data	Looker Studio	—
 [Sample] Rio Olympics Data	Looker Studio	—

Add

The **Report Editor** will open, and a chart connected to your existing data source will already be added to your report.

Untitled Report

File Edit View Insert Page Arrange Resource Help

Reset Share View

Add page Add data Add a chart Add a control Theme and layout

	First Name	Record Count
1.	Paris	51
2.	Norman	49
3.	Sydney	47
4.	Daryl	47
5.	Jamey	47
6.	Loren	45
7.	Leon	45
8.	Casey	44
9.	...	...

1 - 100 / 5162

Chart

SETUP STYLE

Data source

CustomerLoyaltyProgram.csv

+ BLEND DATA

Date Range Dimension

+ Add dimension

Dimension

ABC First Name

+ Add dimension

Drill down

Metric

AUT Record Count

+ Add metric

Optional metrics

Metric sliders

Rows per Page

100

Data

Search

CustomerLoyaltyProgram.csv

- ABC City
- 123 Count
- ABC Country
- ABC Coupon Response
- 123 Customer Lifetime Value
- ABC Customer Name
- ABC Education
- ABC First Name
- ABC Gender
- 123 Income
- ABC Last Name
- 123 Latitude
- ABC Location Code
- 123 Longitude
- 123 Loyalty Count
- 123 Loyalty#
- ABC LoyaltyStatus
- ABC Marital Status
- 123 MonthsAsMember
- 123 Order Year
- ABC Postal code
- ABC Product Line
- ABC Province or State
- 123 Quantity Sold

Adding Visualizations to a Report

You can add various types of charts to your reports to help tell a visual story about your data.

In the **Report Editor** window, click **Add a chart** from the toolbar, then select a chart type from the list, such as a bar chart, pie chart, line chart, map, or bubble chart.

Untitled Report

File Edit View Insert Page Arrange Resource Help

↶ ↷ ⚡ 🔍 Add page 📄 Add data **📊 Add a chart** 🧩 Add a control <

	First Name	Record Count
1.	Paris	51
2.	Norman	49
3.	Sydney	47
4.	Daryl	47
5.	Jamey	47
6.	Loren	45
7.	Leon	45
8.	Casey	44
n	Table	1 - 100 / 5162 < >

Table

Scorecard

Total 1,168 Sessions 69.3K

Time series

Bar

Pie

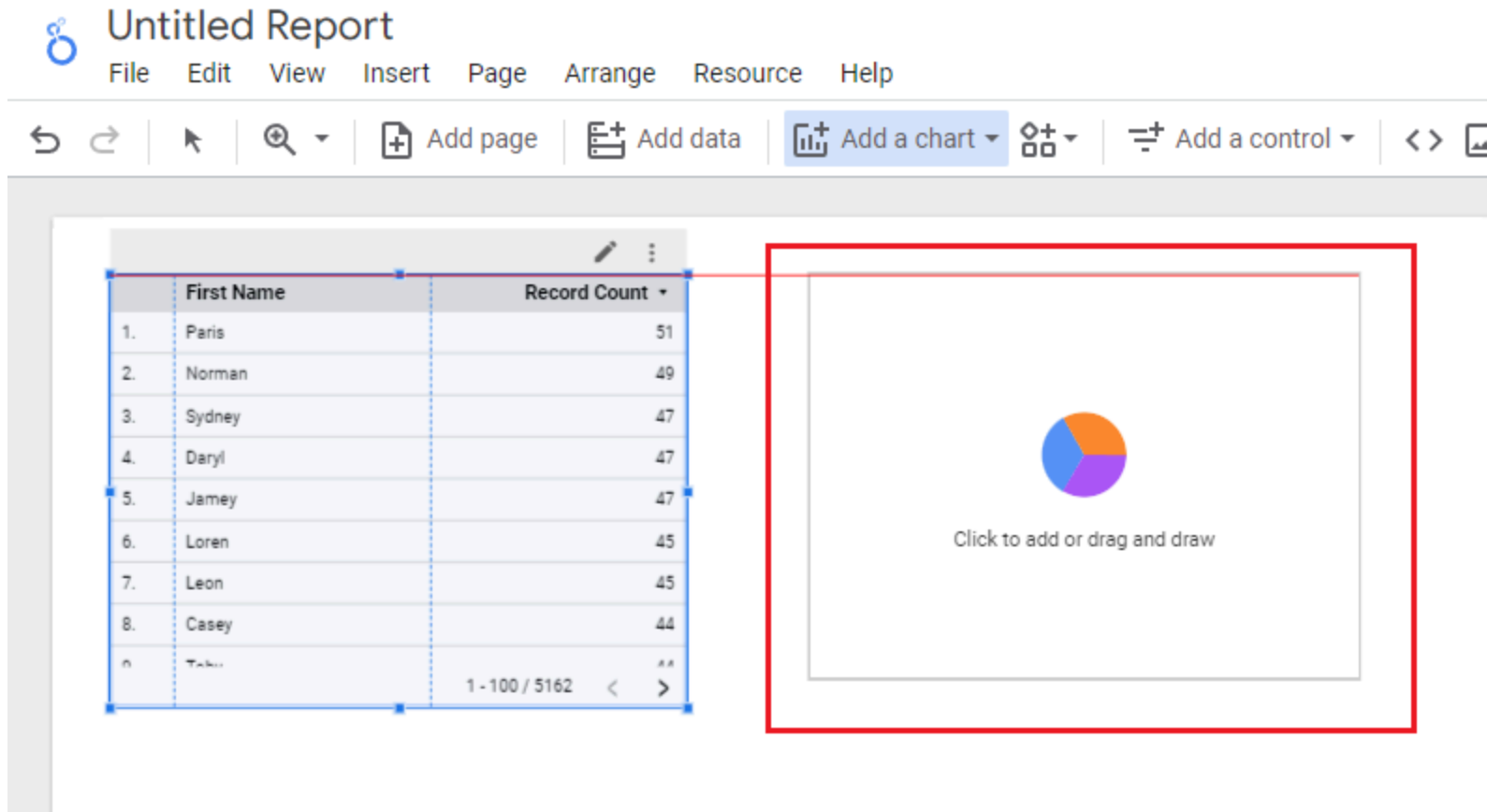
Google Maps

Geo chart

Line

Area

Having selected a chart type, such as a pie chart, you then decide where you want to place the new chart on your canvas by clicking the mouse in that position.

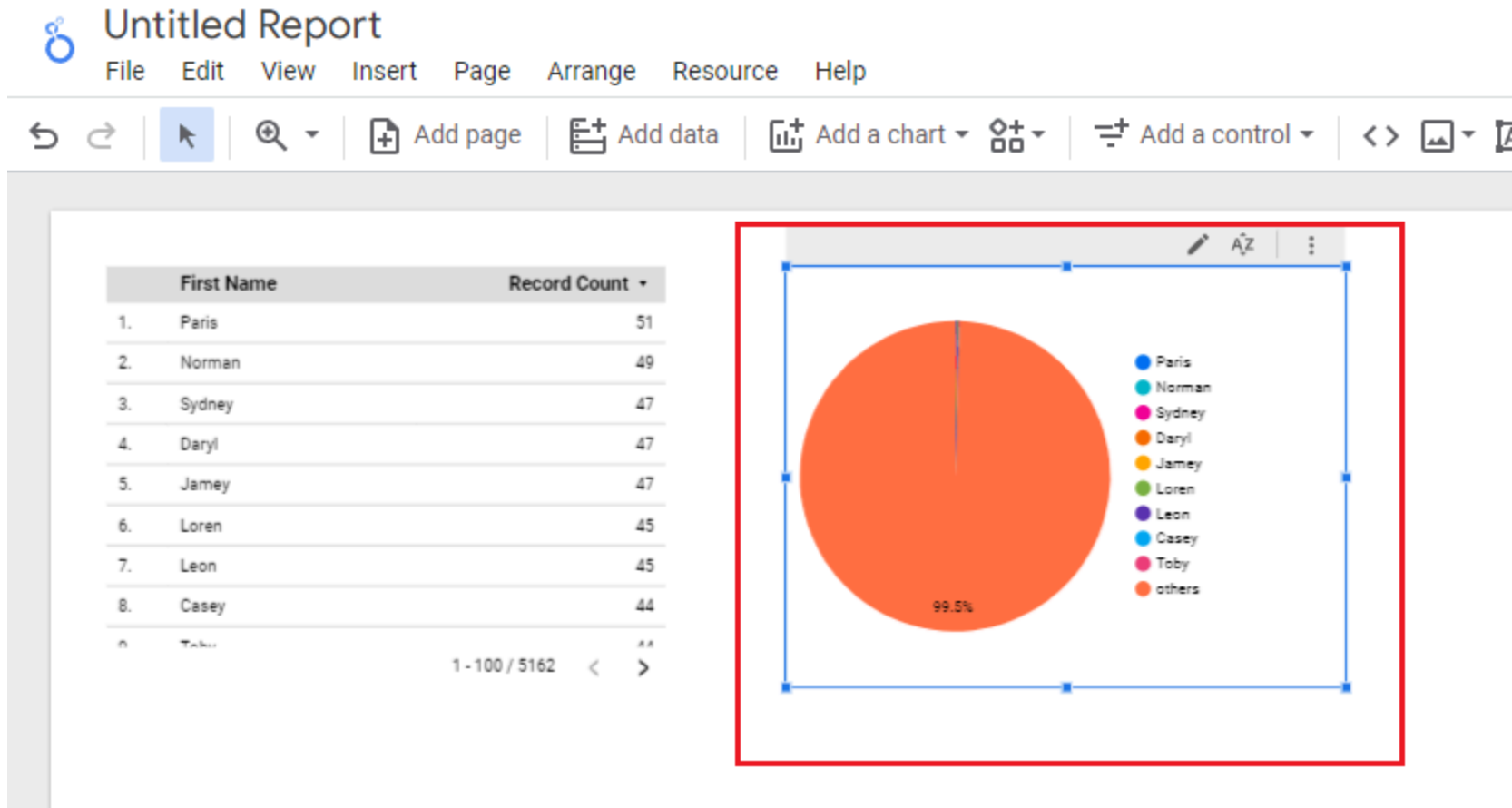


The screenshot shows the Google Looker Studio interface. At the top, there's a menu bar with options: File, Edit, View, Insert, Page, Arrange, Resource, and Help. Below the menu is a toolbar with icons for undo, redo, select, zoom, add page, add data, add a chart (highlighted), add a control, and a full screen icon. The main canvas area is divided into two sections. On the left, there's a table with the following data:

	First Name	Record Count
1.	Paris	51
2.	Norman	49
3.	Sydney	47
4.	Daryl	47
5.	Jamey	47
6.	Loren	45
7.	Leon	45
8.	Casey	44
9.	Tasha	44

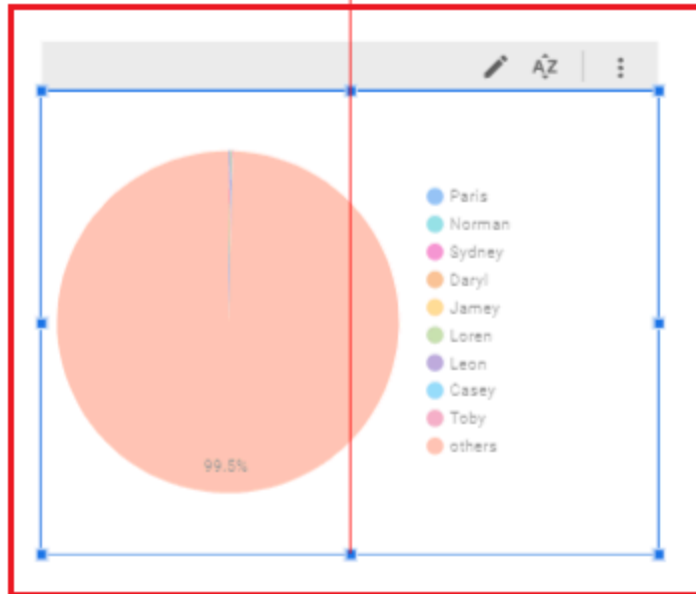
At the bottom of the table, it says "1 - 100 / 5162". On the right side of the canvas, there's a red rectangular box containing a pie chart with three segments (blue, orange, and purple) and the text "Click to add or drag and draw". A red line connects the "Record Count" column header of the table to the pie chart, indicating the data source.

The chart will be added to the canvas.

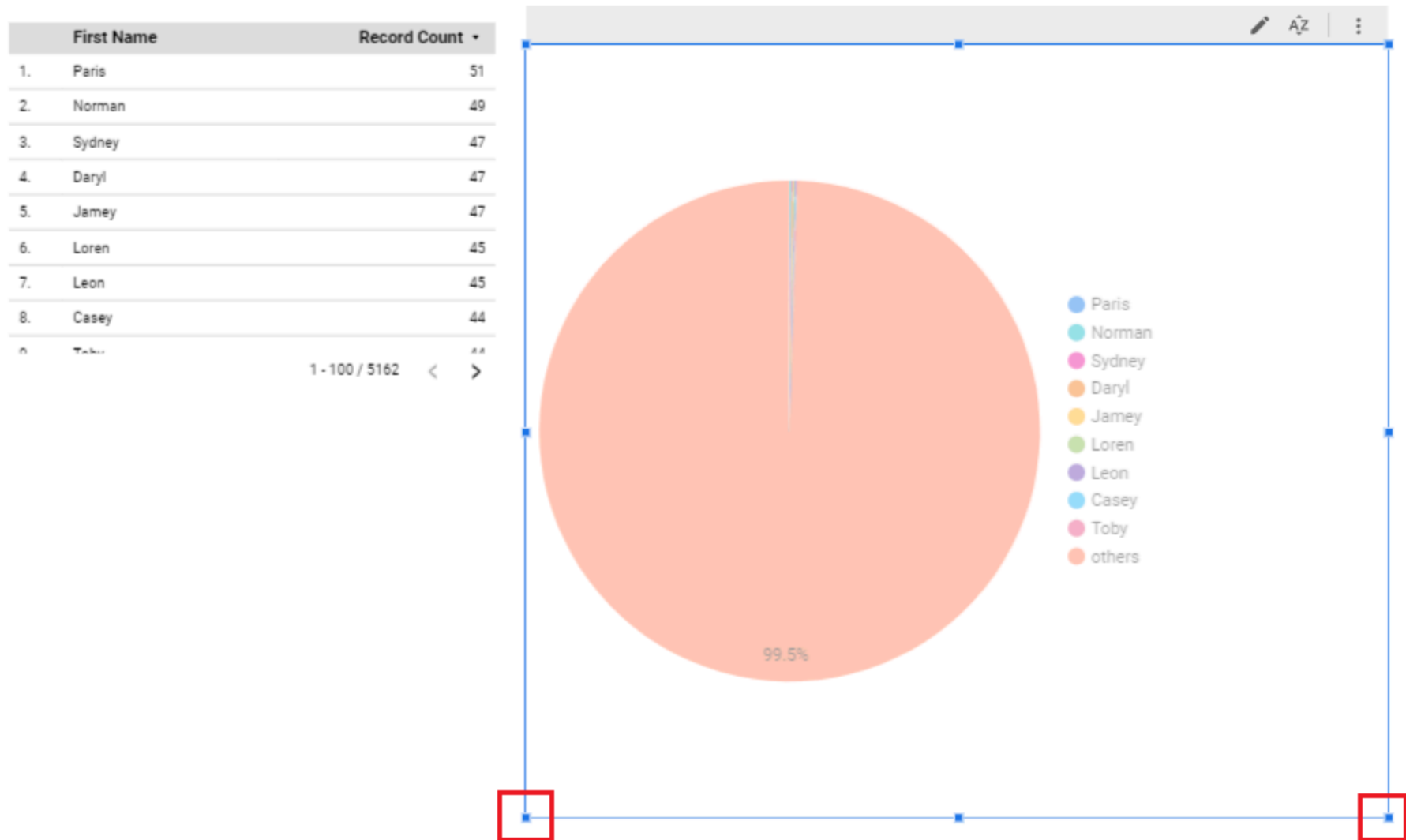


You can drag and drop a chart anywhere else on the canvas if you decide you want to change its position.

	First Name	Record Count ▾
1.	Paris	51
2.	Norman	49
3.	Sydney	47
4.	Daryl	47
5.	Jamey	47
6.	Loren	45
7.	Leon	45
8.	Casey	44
9.	Toby	44
		1 - 100 / 5162 < >

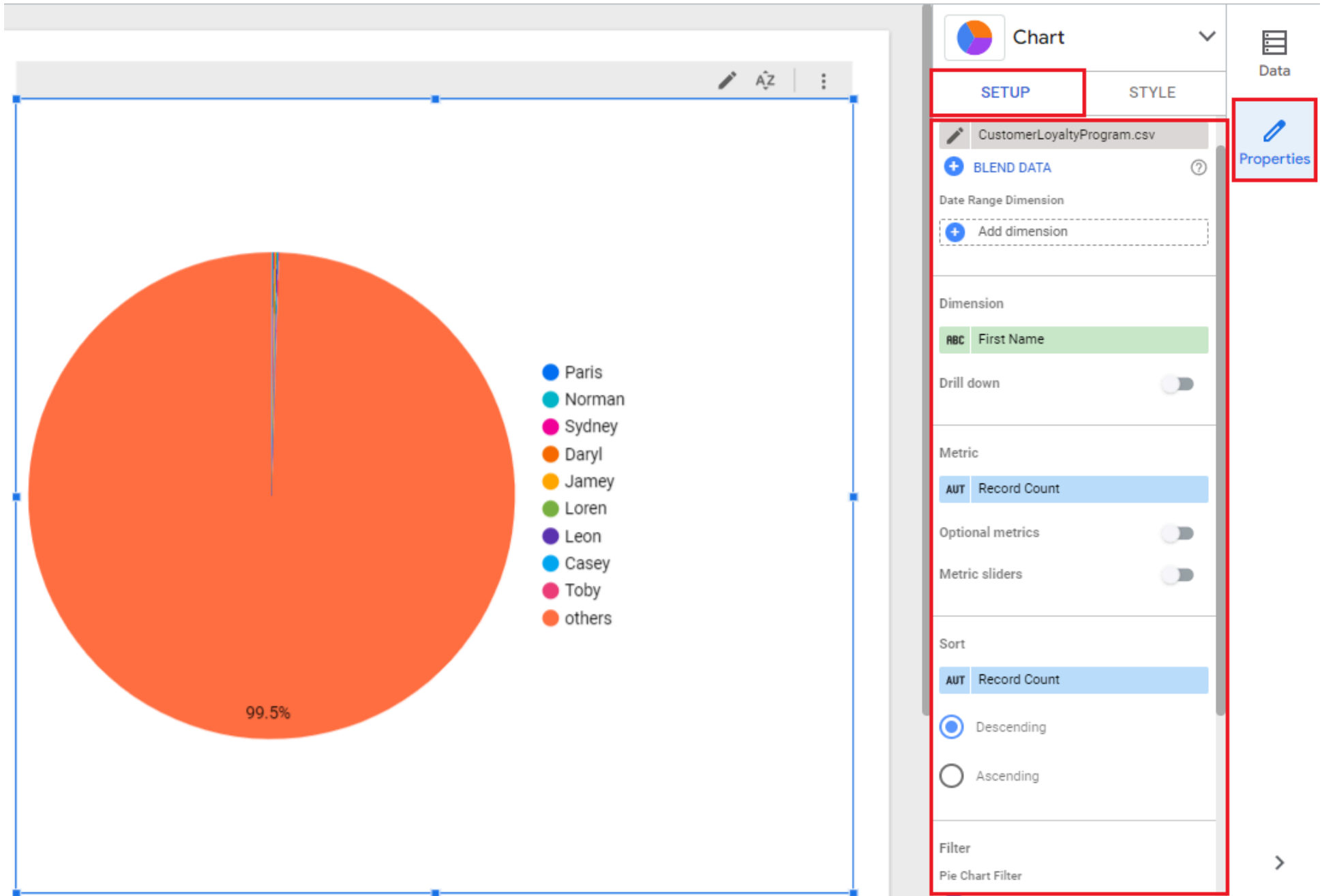


You can use the *drag handles* in the corners to resize the chart object.

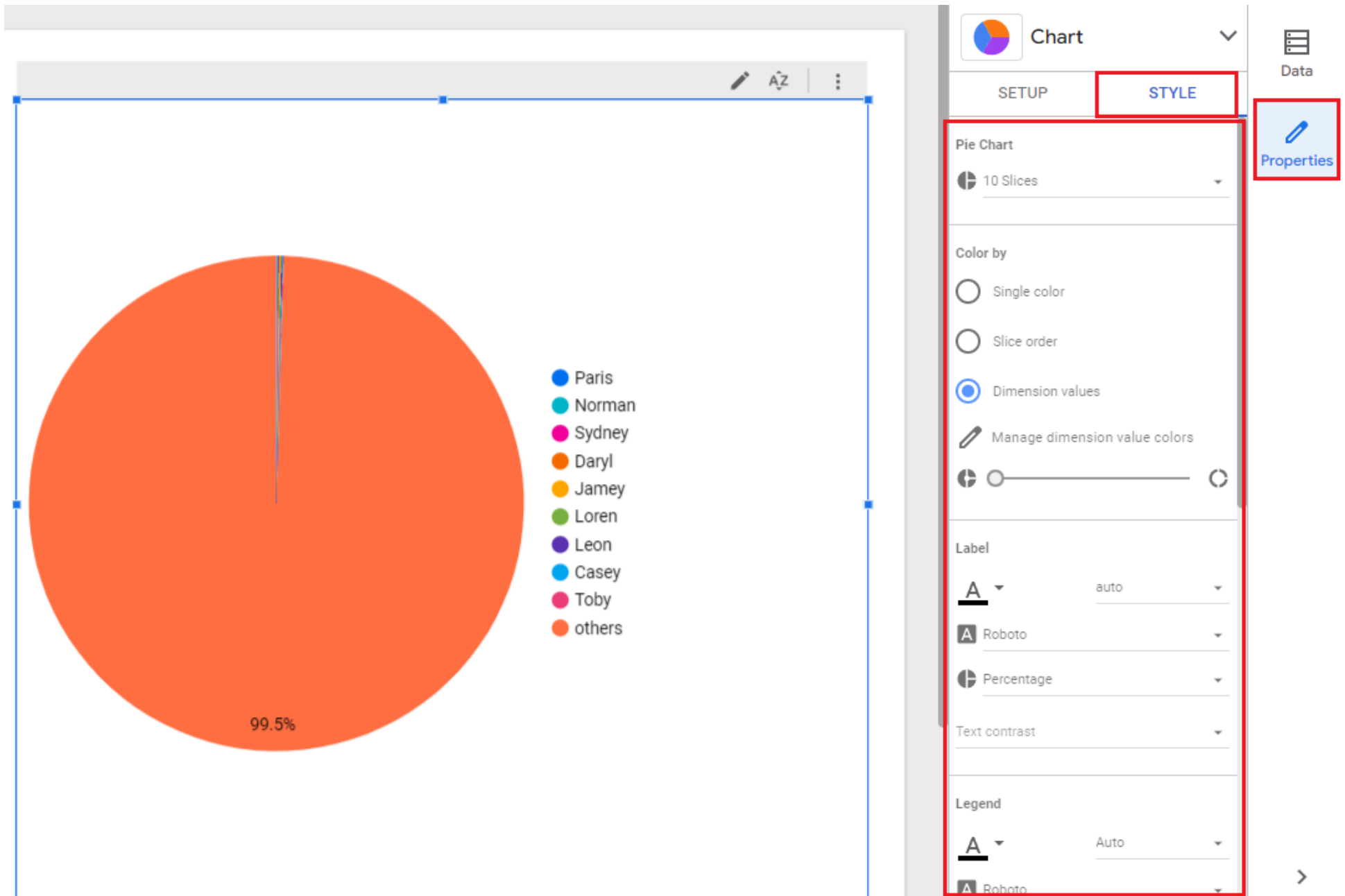


Whenever a chart is selected, you will see a **Data** button and a **Properties** button displayed on the far right side of the page.

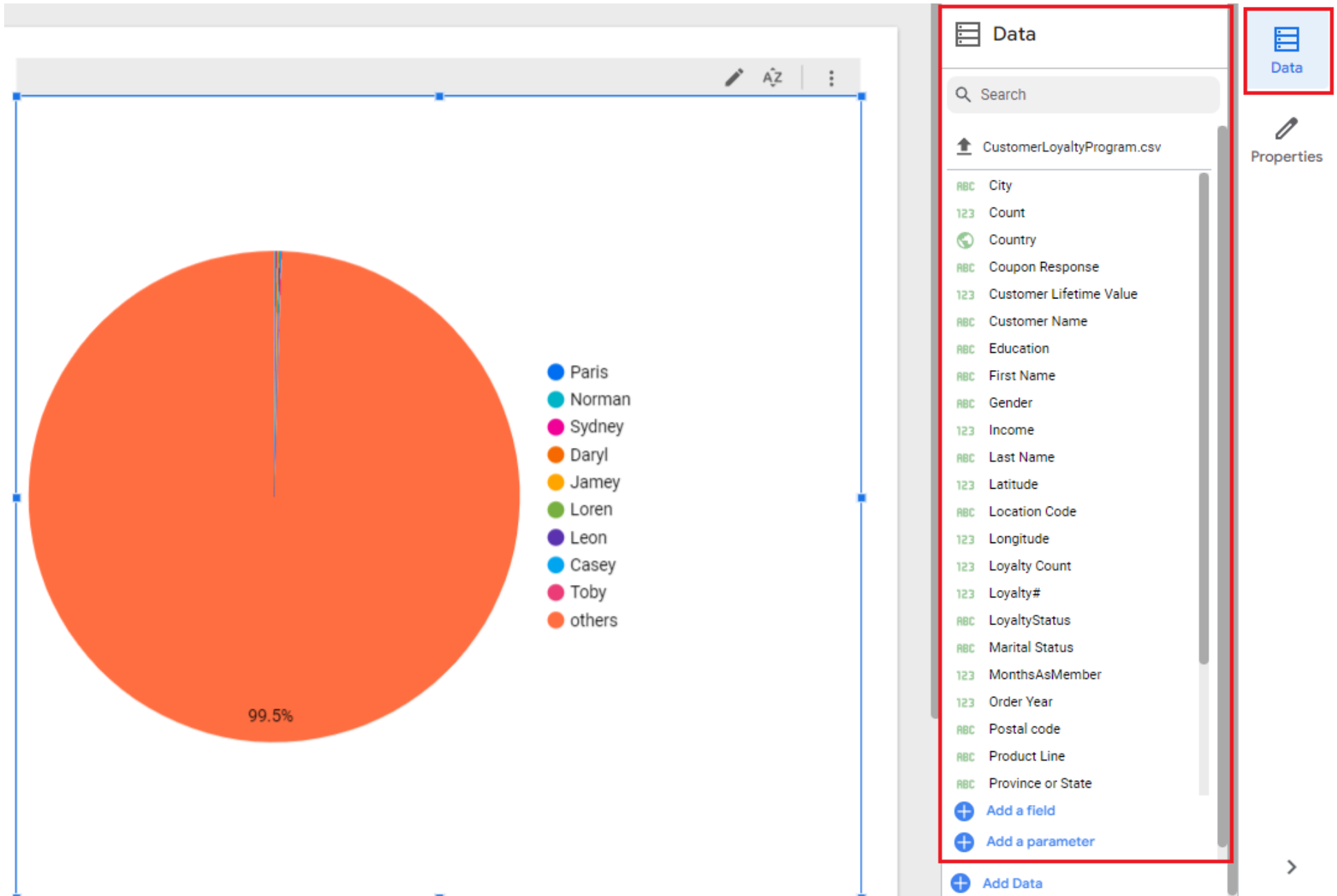
The **Properties** button displays the chart's properties in a pane on the right, and you can use this to configure your chart. On the **SETUP** tab, you configure the data fields that are displayed in the chart and configure filters.



On the **STYLE** tab, you configure the look and feel of the chart, such as the colors used, the data labels, the legend, and the background and border.

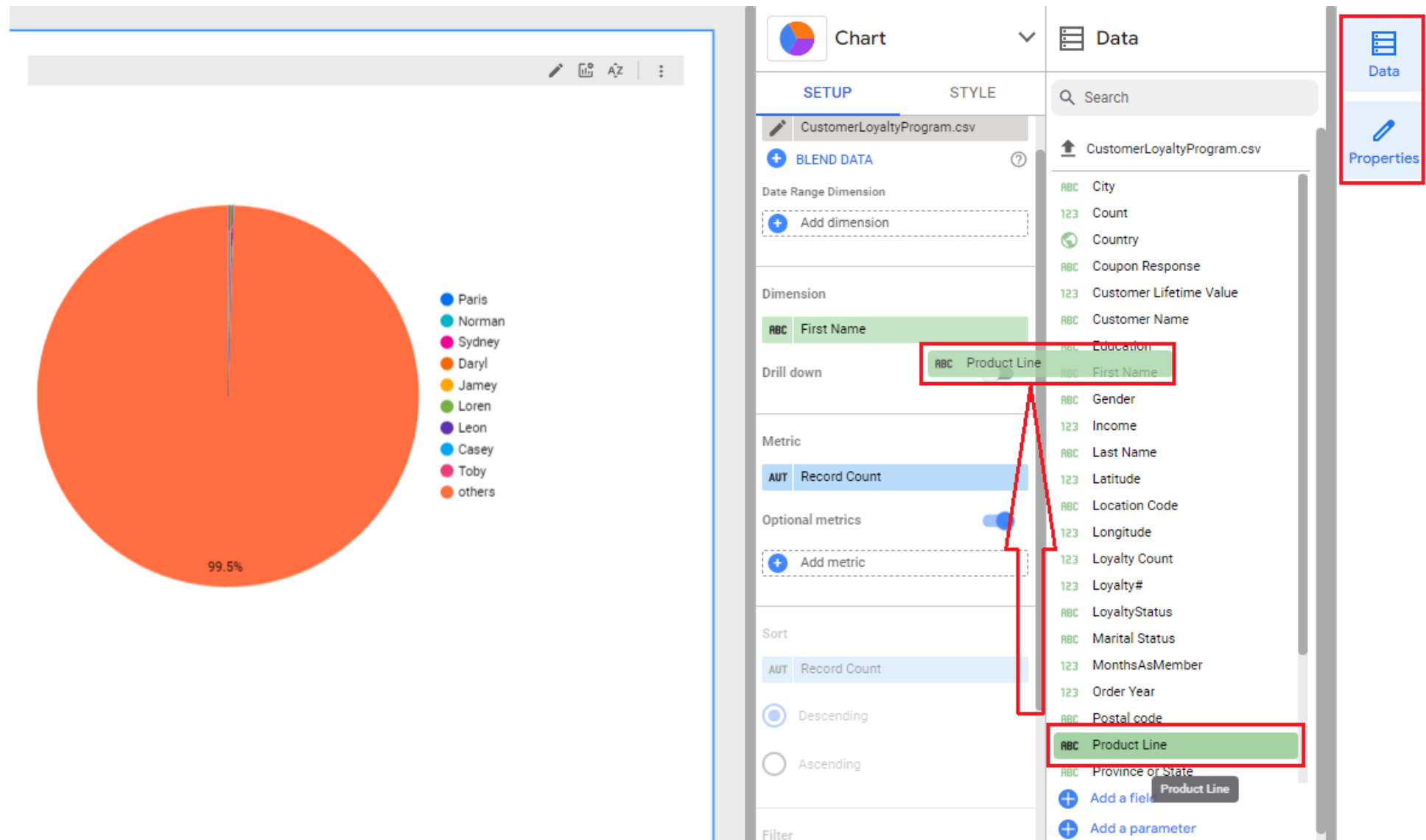


The **Data** button displays the source data embedded in the chart and all the fields contained in that data source.

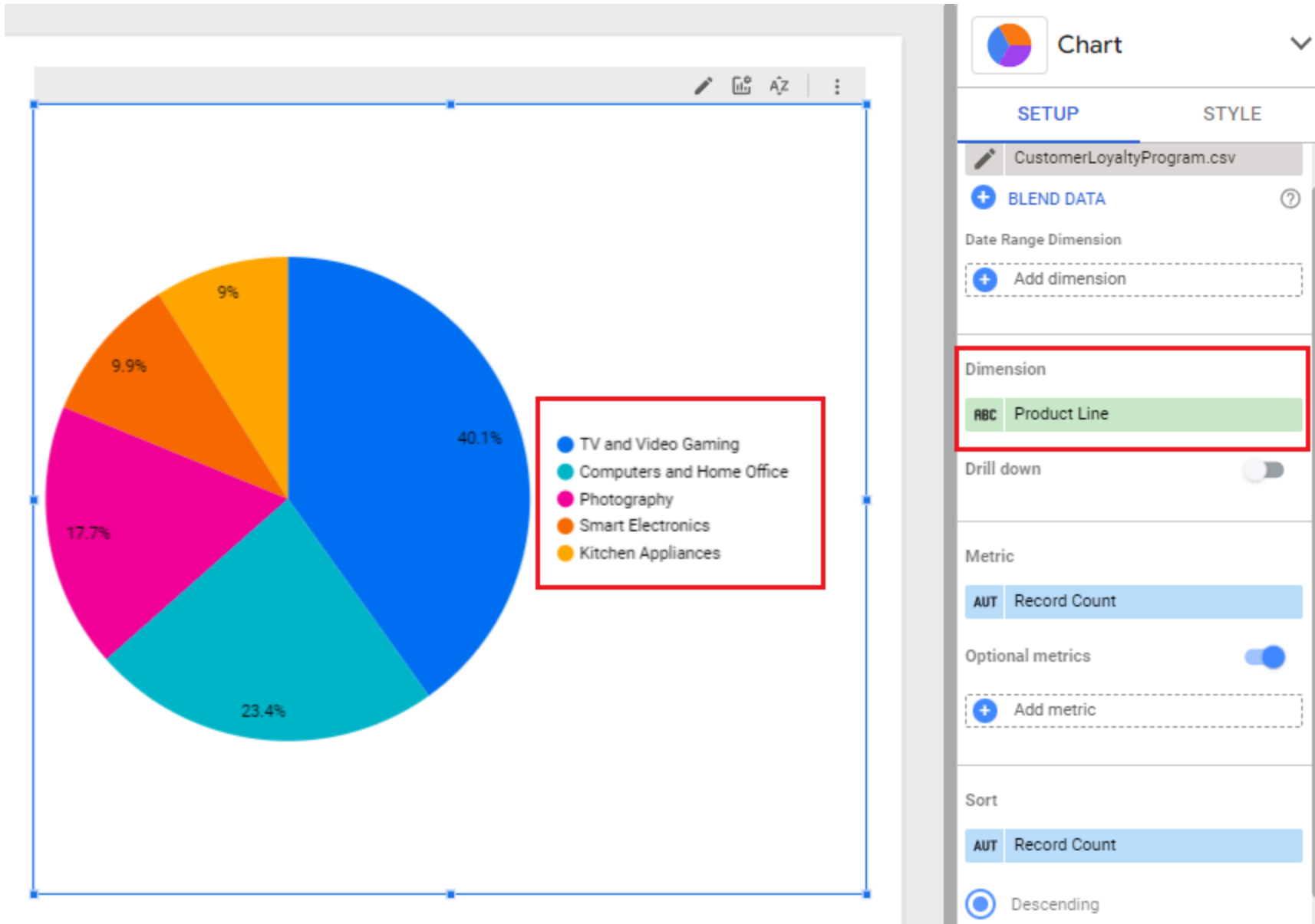


Both these buttons can be toggled on or off to show or hide them.

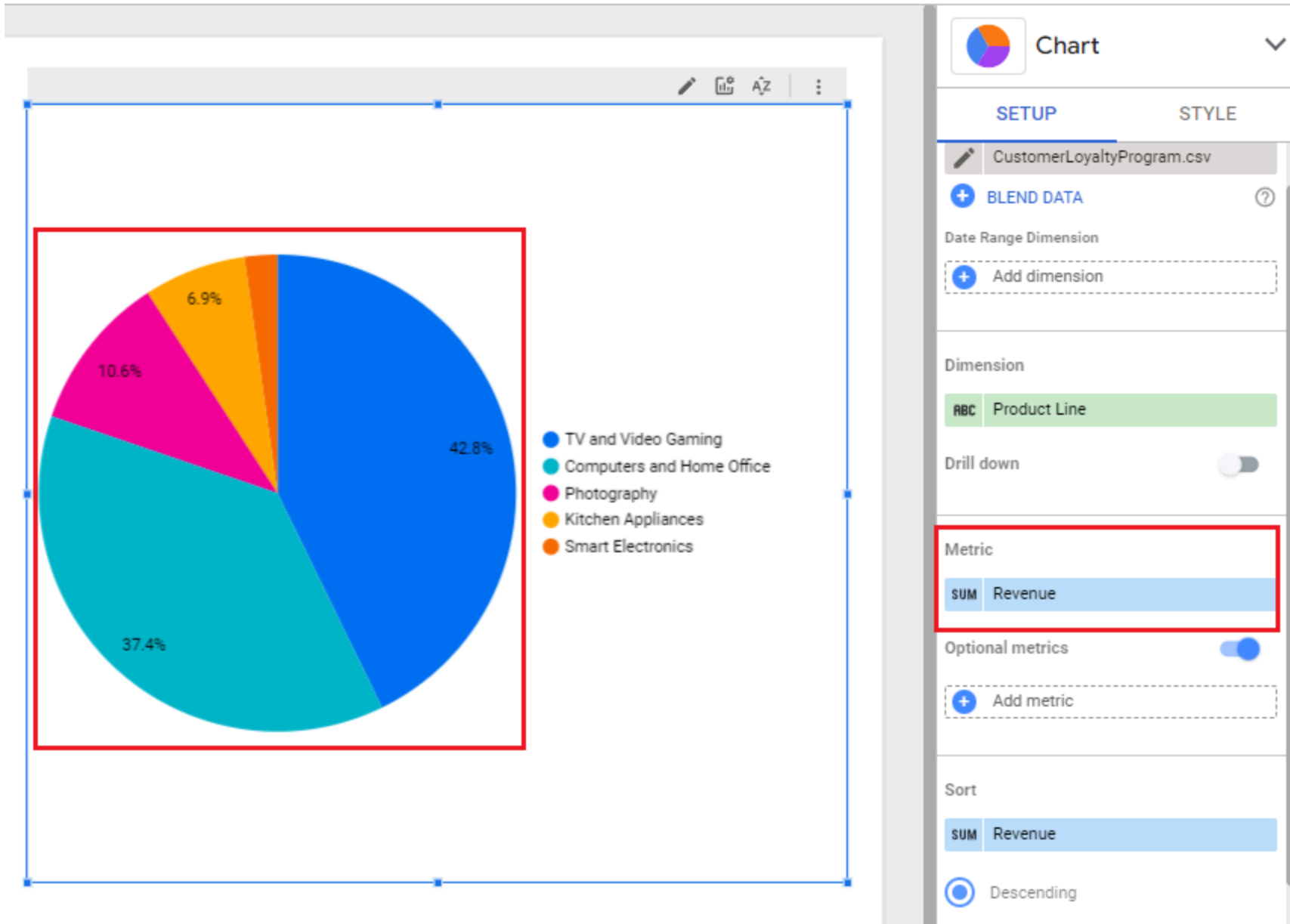
With both buttons switched on, you can easily drop a field from the **Data** pane to a **Dimension** or **Metric** field in the **SETUP** tab of the **Properties** pane to modify the chart.



The chart will update with your changes. Here, the **Dimension** has been changed from **First Name** to **Product Line**.



While here, the **Metric** has been changed from **Record Count** to **Revenue**.



Using this approach, you can add multiple different visualizations to your reports to create an informative and appealing dashboard.



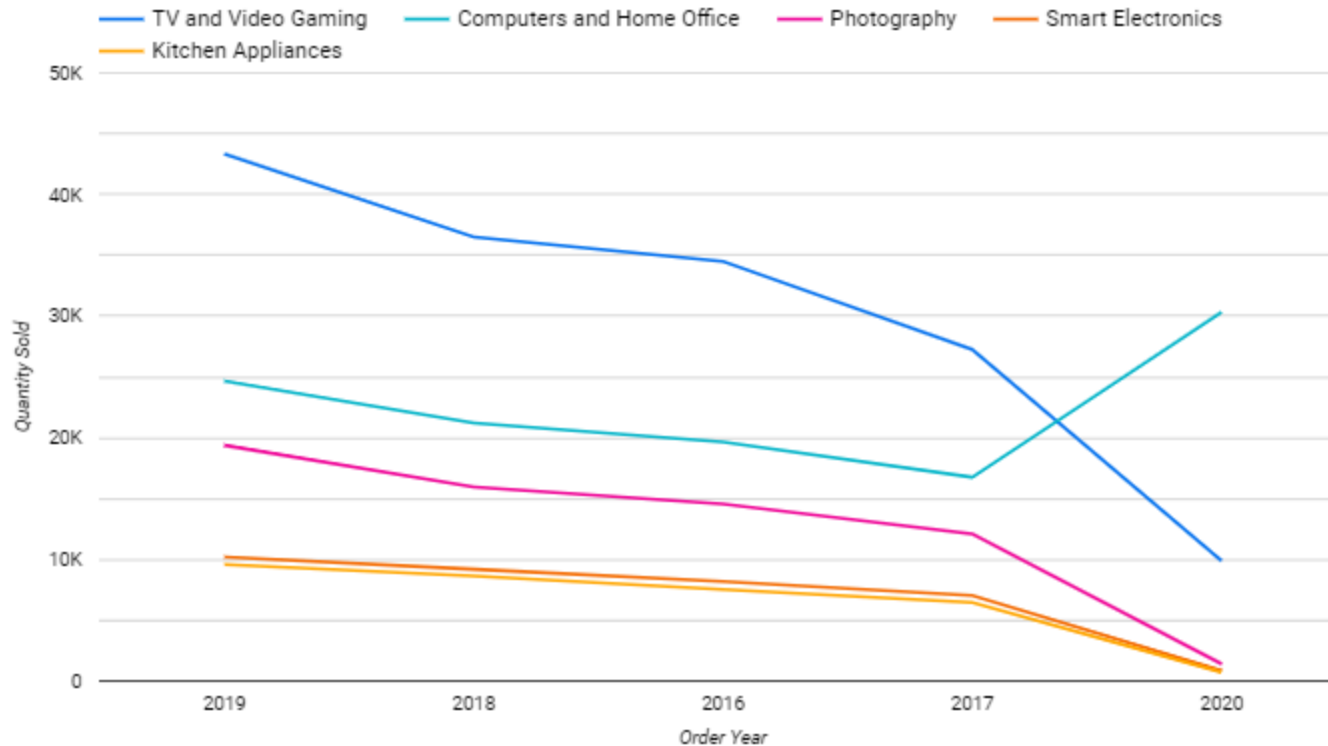
Quantity Sold

396,100

Revenue

228,741,556

Product Line Performance by Year



Configuring Theme and Layout Settings

Looker Studio doesn't use a predefined dashboard template layout for the canvas, as seen in Cognos Analytics by default. This means you have the flexibility to place your visualizations wherever you want on the canvas rather than in specified panels.

What Looker Studio does have is the **Theme and layout** pane. You can open this pane from the **File** menu or toolbar.

Simple Dashboard

File Edit View Insert Page Arrange Resource Help

Share... 1 of 5 → Add data Add a chart Add a control Theme and layout

1 Theme and layout

2 Report settings

3 Version history

4 Publishing settings

5 New report

Make a copy...

Download as

Embed report

Quantity Sold 396,100

Revenue 228,741,556

Product Line Performance by Year

TV and Video Gaming Computers and Home Office Photography Smart Electronics Kitchen Appliances

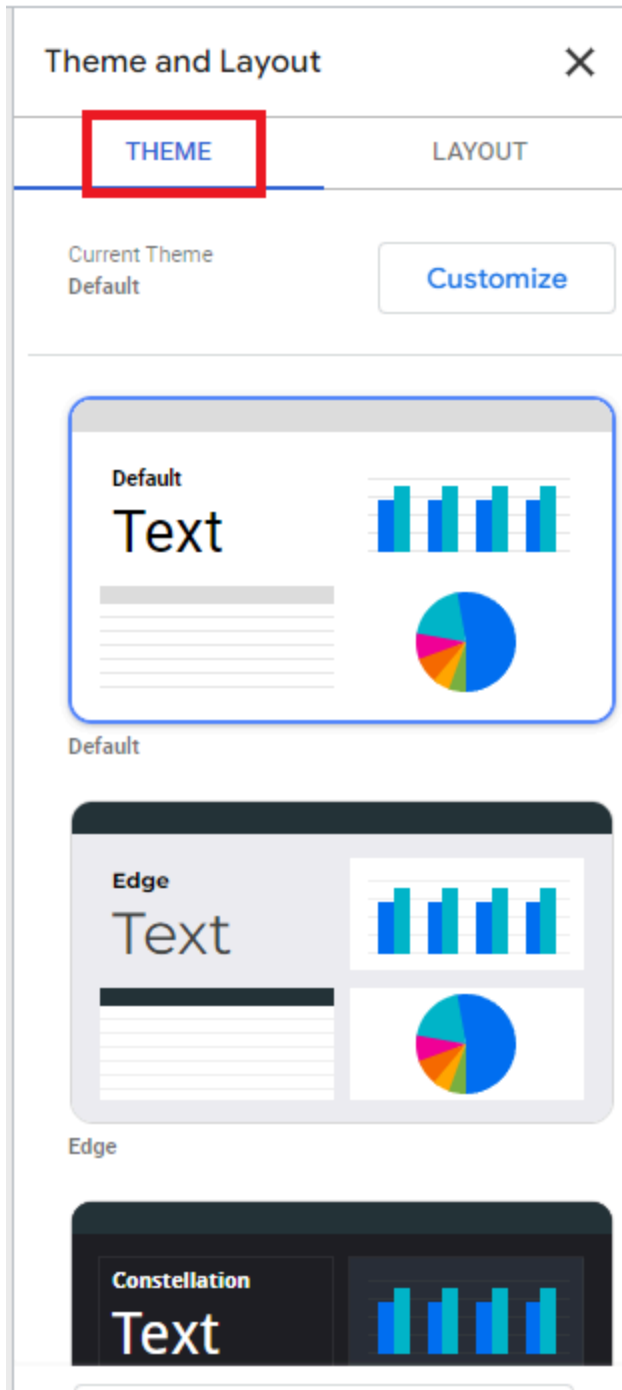
50K

40K

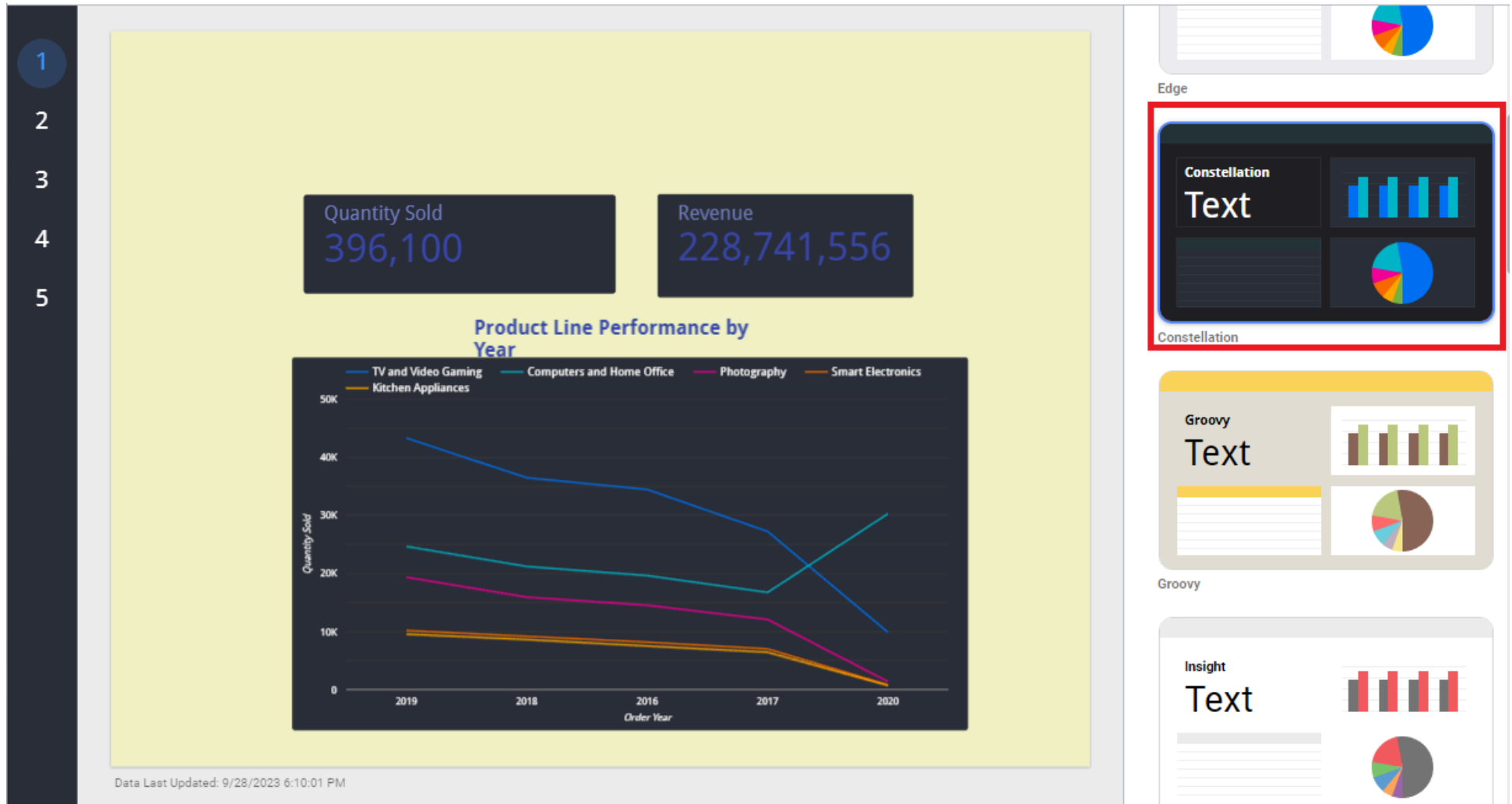
Let's get started

Drag a field from the Data Panel to the canvas to add a new chart or select a component on the report canvas to edit it.

On the **THEME** tab, you can choose one of the predefined themes for your report or make your own custom themes.



When you apply a theme, it gets applied to all the pages in your report.



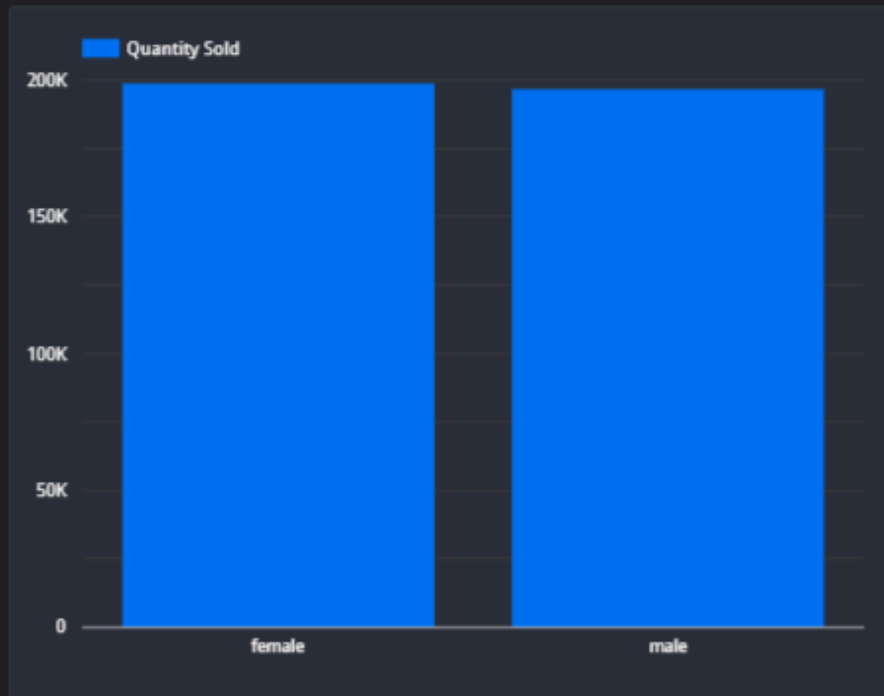
1

2

3

4

5



☒ First Name	Record Count
Type to search	
✓ Paris	51
✓ Norman	49
✓ Sydney	47
✓ Daryl	47
✓ Jamey	47
✓ Loren	45

Gender

Cust_ID

79,436

84,436

Data Last Updated: 9/28/2023 6:12:01 PM

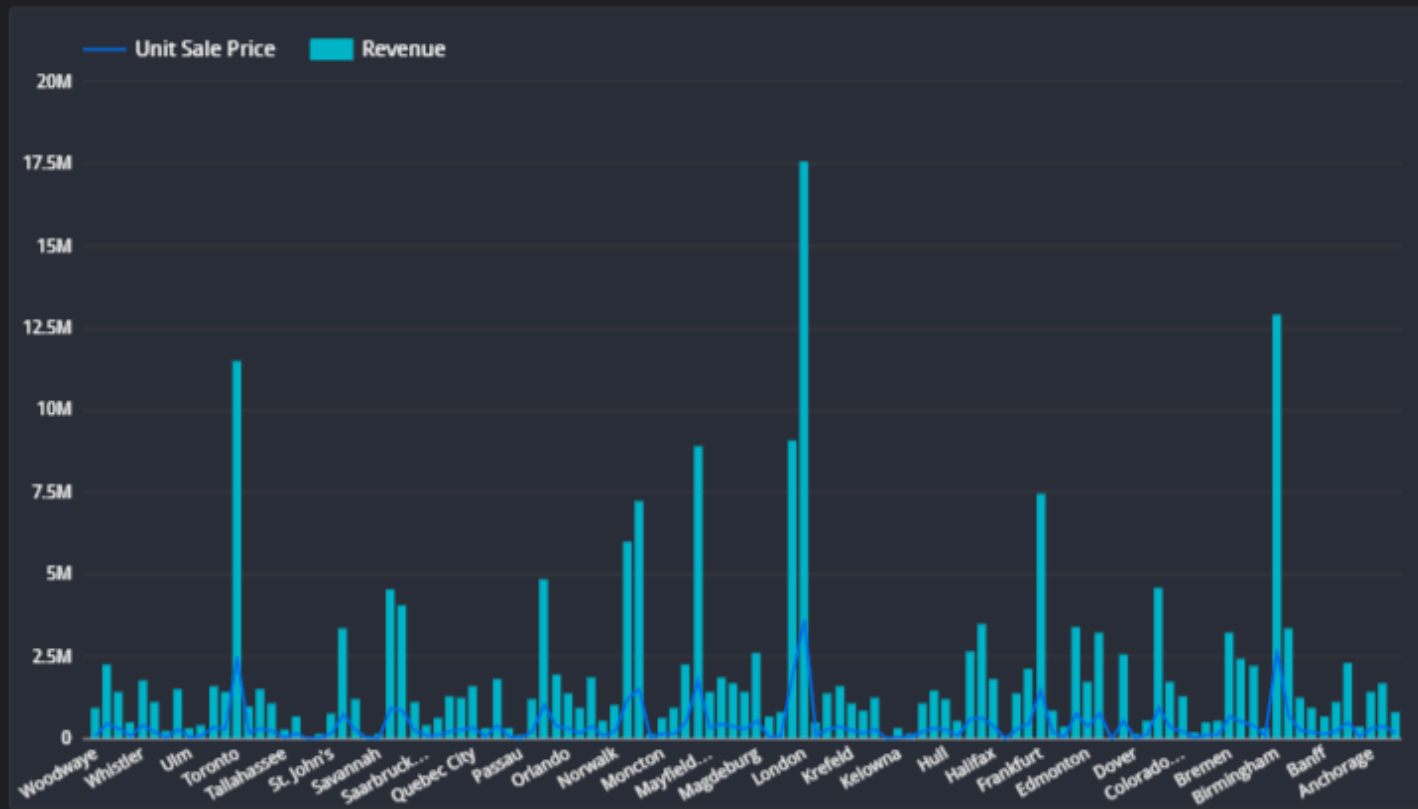
1

2

3

4

5



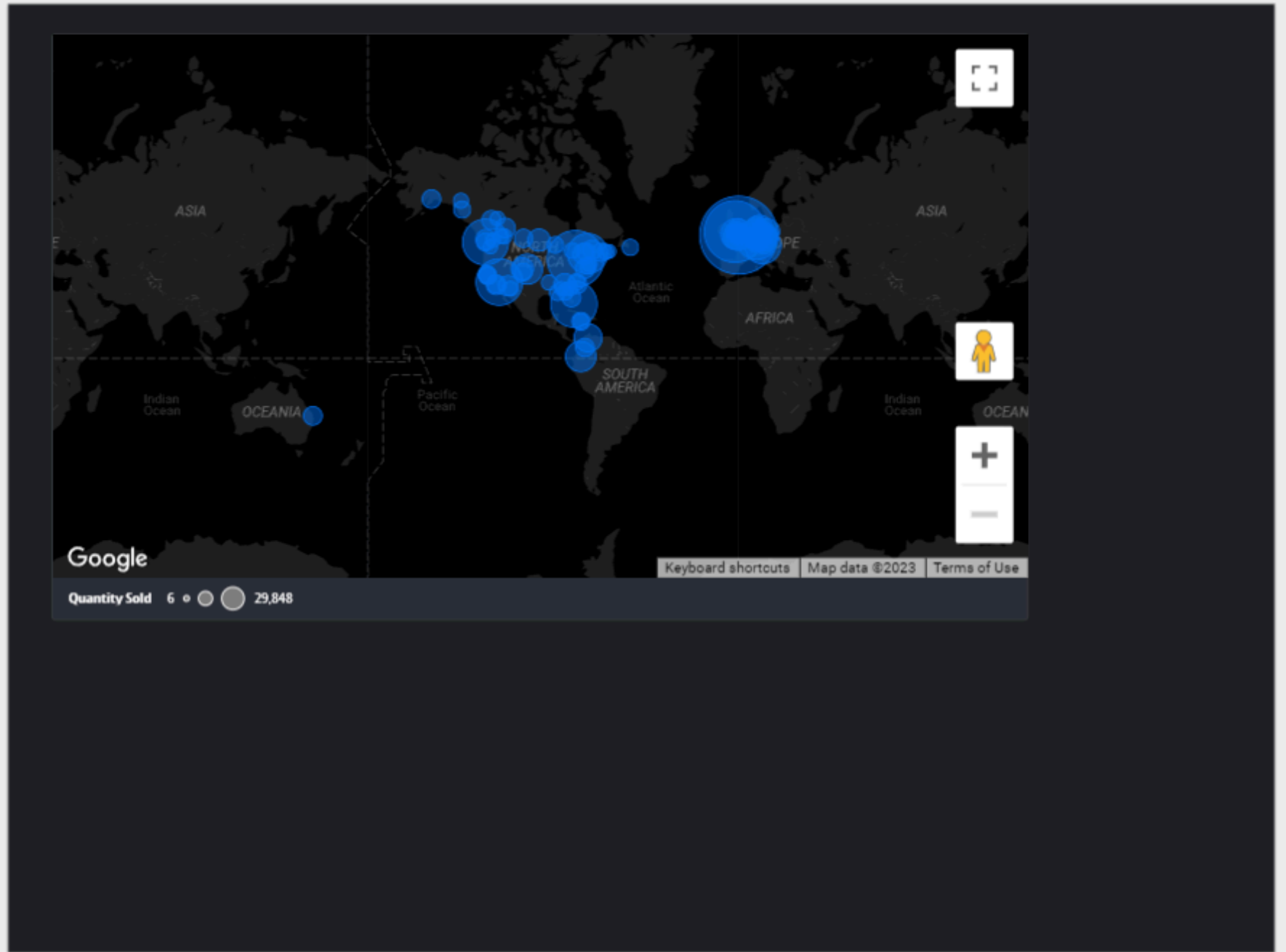
1

2

3

4

5



On the **LAYOUT** tab, you can modify elements of the canvas, such as where the page navigation should be, the display mode, the canvas size, and guides and grid settings.

Theme and Layout

THEME

LAYOUT

View Mode

Header visibility

Always show

Navigation type

Left

☒ Respect report theme

Display mode

☐ Fit to width

☒ Actual size

☒ Has margin

Canvas Size

US letter (4:3) - Landscape

Width (px)

1200

Height (px)

900

Snap to

☒ Smart guides

☐ Grid

✓ **Completed**